Jack Luk

jackluk518@gmail.com | linkedin.com/in/jack-luk | jackluk.com | Vancouver, BC

TECHNICAL SKILLS

Languages: C++, Python, VHDL, MATLAB, Bash
Embedded & Hardware: FPGA (Intel Cyclone V), Arduino, Raspberry Pi, sensor and servo interfacing, soldering
Lab & Debug: Oscilloscopes, bench power supplies, ModelSim simulation, Quartus Prime synthesis
Software & Tools: Linux, Git, Docker, OpenCV, MediaPipe, scikit-learn, NumPy, pandas, SolidWorks

PROJECTS

Vision-Based Rear Monitor
Oct 2026 – Present

Python, OpenCV, Embedded C, CAN bus, Raspberry Pi 5, STM32

- Developing a **two-node embedded system** for cyclists: a Raspberry Pi 5 perception node detects vehicles approaching from behind and transmits threat levels over **CAN bus** to an **STM32** alert unit driving a light and buzzer.
- Architected the CAN message set (detection status, **100 ms heartbeat**) with a **500 ms timeout failsafe** that drives a fault pattern when communication is lost, mirroring automotive ECU practice.

FPGA Digital Alarm Clock
Jul 2026 – Aug 2026

VHDL, Intel DE10-Standard (Cyclone V), Quartus Prime

- Co-developed an **FSM-driven alarm clock** in VHDL with mode control, live timekeeping, and time and alarm setting, synthesized and verified on a Cyclone V FPGA.
- Implemented **debounced, edge-detected** button inputs and **one-hot state encoding** to eliminate spurious transitions from mechanical switch bounce.
- Integrated modular counter, editing, and display subsystems through a **shared control interface**.

Automatic Blueberry Sorter
Sep 2025 – Dec 2025

Arduino, TCS3200 Sensor, SG90 Servo, DC Motors

- **Led the team as project manager**, scheduling tasks and integration milestones to deliver a working electromechanical prototype within a 4-month term.
- Programmed Arduino firmware that reads a TCS3200 sensor and actuates an SG90 servo to **classify blueberries and reject non-conforming objects in real time**.
- Engineered a **motor-driven centrifugal mechanism** that automates physical separation of sorted fruit.

AI Posture Detection System
Nov 2022 – Apr 2023

Python, OpenCV, MediaPipe, scikit-learn

- Independently built a **real-time webcam application** that classifies posture and alerts the user, training an image classification model on self-collected data processed with pandas.
- Selected to advance to the **Greater Vancouver Regional Science Fair**.

EXPERIENCE & LEADERSHIP

Tech Team Member, *SFU Racerbot Club* — Oct 2026 – Present

- Selected for the Tech Team of a student club that designs, builds and races **1/10-scale autonomous cars**.
- Work on the car's **mechanical, electrical and fabrication** tasks and its **Ubuntu/NVIDIA Jetson** compute platform.

Youth Leader, *33rd Richmond Scouts (Richmond, BC)* — Sep 2020 – Present

- Coordinate weekly programs and lead large-scale fairs and camps across **6 years** of continuous volunteer leadership.

First Year Lead, *SFU Canadianized Asian Club (Burnaby, BC)* — Oct 2025 – Aug 2026

- Organized events and fundraisers that engaged SFU's Asian student community.

Private STEM Tutor, *STEM Power to Learn (Remote)* — Oct 2023 – Jun 2025

- Planned and delivered weekly online math sessions for elementary students over **2 academic years**.

EDUCATION

Simon Fraser University, Burnaby, BC — Anticipated May 2029

Bachelor of Applied Science, Systems Engineering

Relevant coursework: Electric Circuits, C++ Programming, Digital Logic Design (VHDL), Engineering Graphics and CAD